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Internal Assessment

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The relationship between the surface area-to-mass ratio and the final vertical velocity of a falling spherical object.

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1. Research Design

1.1 Introduction

In our day-to-day lives, we occasionally drop objects. When this occurs, I notice that different objects touch the ground at various times. I realized that this is due to the difference in velocities between each object, which may be affected by the properties of the object itself. For this Physics IA, I decided to investigate the effect of the surface area-to-mass ratio of a spherical object on its final vertical velocity, recorded 1 meter from the ground. I will be investigating both theoretical and experimental values.

This investigation has real-world significance in fields where the motion of spherical objects through air must be optimized. In sports, the design of balls in soccer, tennis, golf, and baseball relies on carefully balancing mass and surface area to control speed, trajectory, and safety. In engineering, these principles are applied in the development of parachutes and safety equipment, where maximizing drag reduces impact velocity. By modeling how different spheres fall under drag, this experiment connects to broader applications in sports, safety designs, and aerodynamics.

1.2 Research Question

How does the surface area-to-mass ratio (m^2kg^{-1}) of different spherical objects (0.15, 0.20, 0.30, 0.40, 0.55) affect their final vertical velocity (ms^{-1}) one meter from the ground when dropped from a constant height as measured by Tracker Video Analysis?

1.3 Background Information

1.3.1 Gravitational and Drag Force

When objects fall in the air, they experience 2 types of forces. The first force is the gravitational force, which pulls objects toward the center of the planet they are on, in this case, Earth (Tsokos, 2023, 32). The second force is the drag force, which opposes the motion of the object in a fluid, which in this case is air (Tsokos,

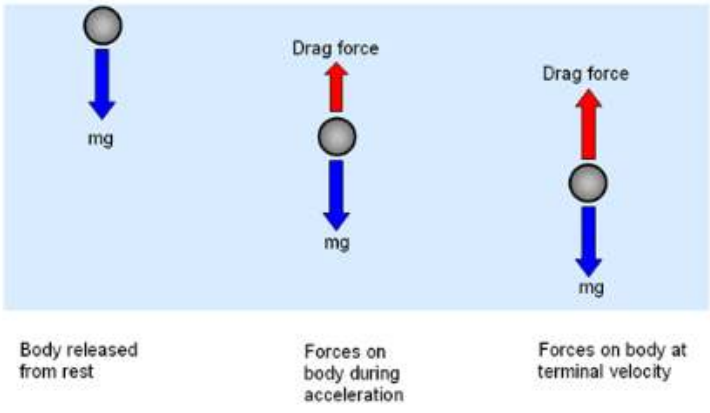


Figure 1: Diagram of a spherical object affected by gravitational and drag force (Wise, n.d.)

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2023, 32). This is illustrated in **Figure 1**.

At first, the object will accelerate, and as it gets faster, it will also experience a greater drag force. Eventually, both forces will cancel out to the point where their acceleration is zero, meaning the object is in terminal velocity. The formula for gravitational force is expressed as:

$$F_g = mg \tag{1}$$

Where:

- F_g represents the gravitational force experienced by the object (N)
- m represents the mass of the spherical object (kg)
- g represents the gravitational field strength (m/s^2)

On the other hand, the drag force is defined as:

$$F_d = \frac{1}{2}\rho C_d A v^2 \tag{2}$$

Where:

- F_d represents the drag force experienced by the object (N)
- ρ represents the fluid density (kg/m^3)
- C_d represents the drag coefficient of a sphere (dimensionless)
- A represents the sphere's cross-sectional area (m^2)
- v represents the velocity of the object (m/s)

It can also be noticed that the drag force uses the sphere's cross-sectional area, which is essentially the area of the object facing the fluid. This will always be a circle, shown in **Figure 2**.

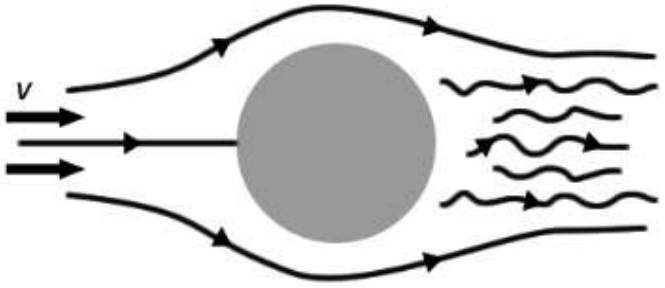


Figure 2: Drag force on a spherical object (SMU, 2009)

If we let the radius of the sphere be r, then its cross-sectional area would be:

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$$A = \pi r^2 \tag{3}$$

Where:

- r represents the sphere's radius (m)

Substituting this back into **Equation (2)**, we get the formula of the drag force to be:

$$F_d = \frac{1}{2}\rho C_d \pi r^2 v^2 \tag{4}$$

1.3.2 Surface Area-To-Mass Ratio

The surface area-to-mass ratio (k) is used as an independent variable because it directly affects how much air resistance (drag force) an object experiences relative to its weight. Using this ratio allows me to integrate two of the most fundamental properties that differ between different objects, making it more applicable to other objects in the real world that are spherical. It can be expressed as:

$$k = \frac{A_s}{m} \tag{5}$$

Where:

- k represents the surface area-to-mass ratio (m^2/kg)
- A_s represents the surface area of the spherical object (m^2)
- m represents the mass of the spherical object (kg)

We know that for a spherical object, its surface area can be calculated by:

$$A_s = 4\pi r^2 \tag{6}$$

Thus, substituting this for **Equation (5)**, we get the final equation to be:

$$k = 4\frac{\pi r^2}{m} \tag{7}$$

1.3.3 Terminal Velocity

As stated previously, when both forces acting on the object equalize, the object will fall at its terminal velocity. Now, I will try to derive terminal velocity as it correlates with the finding of the final vertical velocity as a function of height. In terminal velocity, the gravitational force and drag force equalize, hence:

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$$mg = \frac{1}{2}\rho C_d \pi r^2 v_t^2 \tag{8}$$

By rearranging the equation and substituting **Equation (8)** into m, we get:

$$v_t = \sqrt{\frac{8g}{\rho C_d k}} \tag{9}$$

With this, terminal velocity is inversely proportional to the square root of the surface area-to-mass ratio.

1.3.4 Final Vertical Velocity

Now, I am to find the final vertical velocity (v_f) as a function of height (x), since I cannot guarantee that the objects will attain terminal velocity. We are first to define the drag force in terms of v_t. By rearranging **Equation (8)** and substituting back into the drag-force equation in **Equation (2)**, we get:

$$F_d = (mg)\frac{v^2}{v_t^2} \tag{10}$$

Knowing this, we can substitute **Equation (10)** into Newton's second law:

$$ma = mg(1 - \frac{v^2}{v_t^2}) \tag{11}$$

Since:

$$\frac{dv}{dt} = \frac{dv}{dx} * \frac{dx}{dt} = v\frac{dv}{dx}$$

Substituting this into **Equation (11)** gives us:

$$\frac{vdv}{1-\frac{v^2}{v_t^2}} = gdx$$

I made the equation into this form as I would like to change v it as a function of displacement. Solving the seperable differential equation:

$$\int \frac{vdv}{1-\frac{v^2}{v_t^2}} = \int gdx$$

$$-\frac{v_t^2}{2} ln(1 - \frac{v^2}{v_t^2}) = gx + C \tag{12}$$

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To find this constant, we use the fact that at $x = 0$, $v = 0$ (as the object starts from rest). Substituting these values into **Equation (12)**:

$$-\frac{v_t^2}{2} ln(1) = C$$

$$C = 0$$

Therefore:

$$v = v_t \sqrt{1 - e^{-\frac{2gx}{v_t^2}}}$$

Where v can be denoted as v_f. Finally, substituting the equation of v_t, **Equation (9)**, we get the final function:

$$v_f = \sqrt{\frac{8g}{\rho C_d k}} * \sqrt{1 - e^{-\frac{\rho C_d k x}{4}}} \tag{13}$$

With the derived **Equation (13)**, I can calculate the theoretical final velocity of the object when it is 1 meter away from the ground.

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1.4 Hypothesis

- Null Hypothesis: There is no significant relationship between k and v_f of the spherical objects when dropped from a constant height.
- Alternative Hypothesis: The v_f of spherical objects will be inversely related to the square root of k when dropped from a constant height.

1.5 Variables

Variables	Description	Apparatus
Independent	**Mass** Varied between objects to produce a different surface area-to-mass ratio.	**Digital Scale** The uncertainty is seen from the smallest division of my digital scale.
	Surface Area Varied between objects to produce a different surface area-to-mass ratio alongside the mass.	**Measuring Tape** The uncertainty is seen from the smallest division of the measurement tape.
Dependent	**Final Horizontal Velocity** Measured by tracking the displacement of the object as it falls per frame.	**Tracker Video Analysis Software** The uncertainty is due to the precision of the Tracker Video Analysis Software.

Table 1: Independent and dependent variables

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Controlled Variables	
Variable	**Means of Control**
Shape of object	Prevents varying drag coefficients. Controlled by selecting only spherical objects.
Height of Drop	The height must be constant to prevent anomalous data, being 4.500 meters, controlled by measuring height before every trial.
Drag Coefficient, Density of Air, and Gravitational Acceleration	All coefficients used for the theoretical model must be constant for uniformity. Take $C_d = 0.47$ (Benson, n.d.) $\rho = 1.29\ kgm^{-3}$ (Shah), $g = 9.81\ ms^{-2}$ (National Science Foundation, n.d.)
Calibration Stick	The reference calibration stick is kept constant to prevent systematic errors. The position and length of the 1-meter ruler are not changed in real life and in the Tracker software.
Ball Tracking Method	The same reference point on the object must be tracked to avoid parallax or shape-related inconsistencies. The objects are tracked only from their center of mass.
Object Dropping Method	The object must be released consistently to minimize spin and initial velocity. Each object is placed on a thin cloth held by two strings, and the fabric is pulled vertically upward to allow the object to minimize frictional contact and spin.
Experimental Setup	The setup for each trial must be maintained. The position of each apparatus is not changed per trial.

Table 2: Controlled variables

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1.6 Materials & Apparatus

The materials used for the experiment are as follows:

1. PVC Pipe (5 pcs)
2. Cloth (1 pcs)
3. Baseball (1 pcs)
4. Tennis Ball (1 pcs)
5. Basketball (1 pcs)
6. Soccerball (1 pcs)
7. Volleyball (1 pcs)
8. String

Meanwhile, the apparatus used for the experiment is as follows:

1. Digital scale (1 pcs) (±0.00001 kg)
2. Vernier caliper (1 pcs) (±0.00025 m)
3. Measuring tape (1 pcs) (±0.0005 m)
4. 1 Meter Ruler (1 pcs) (±0.0005 m)
5. Clamp
6. Tripod (1 pcs)
7. Phone camera (1 pcs) (60 fps, HD mode)
8. Laser Level

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1.7 Methodology & Experiment

1.7.1 Figures & Setup

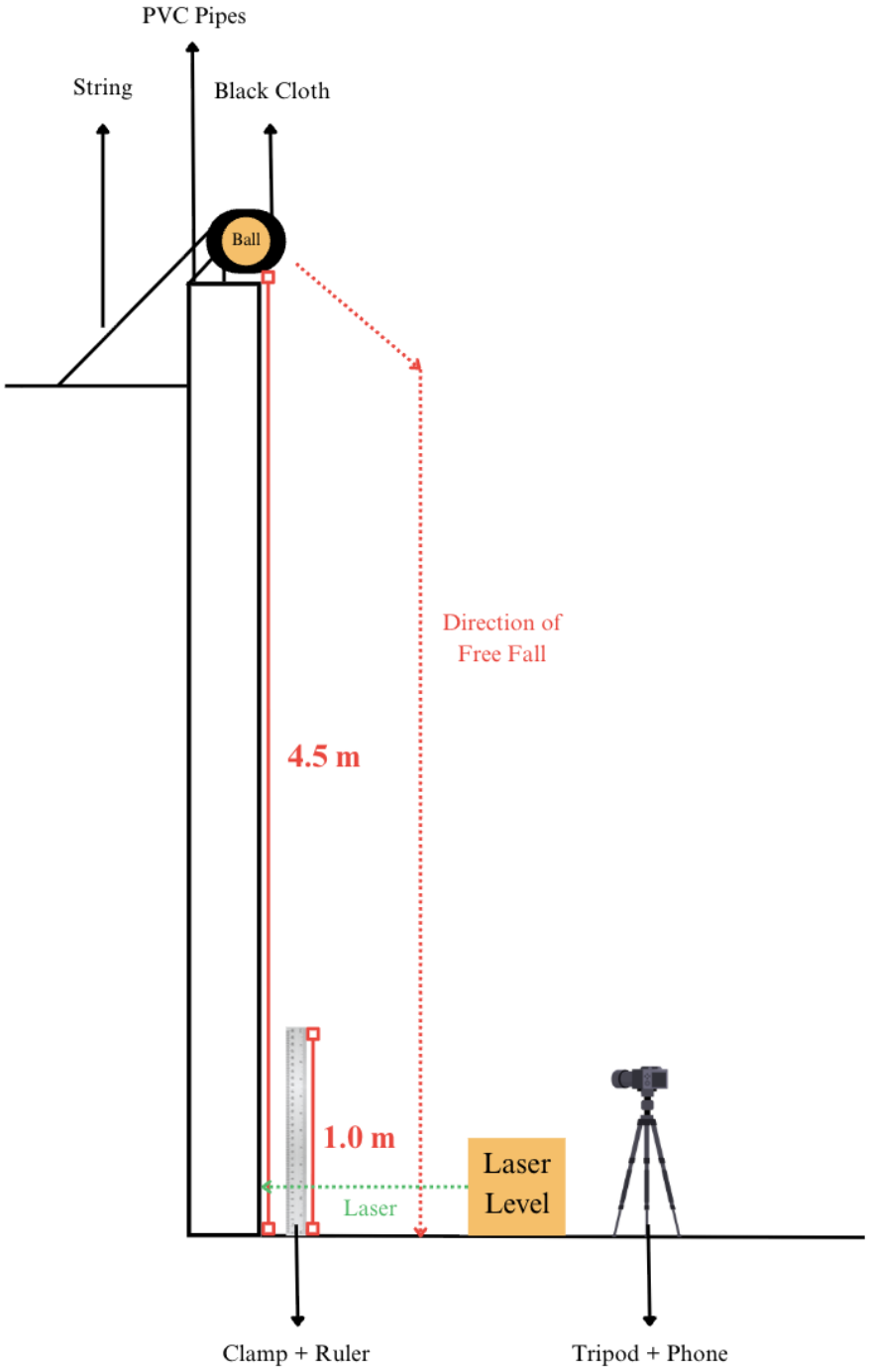


Figure 3: Experimental setup of the experiment (designed with Canva)

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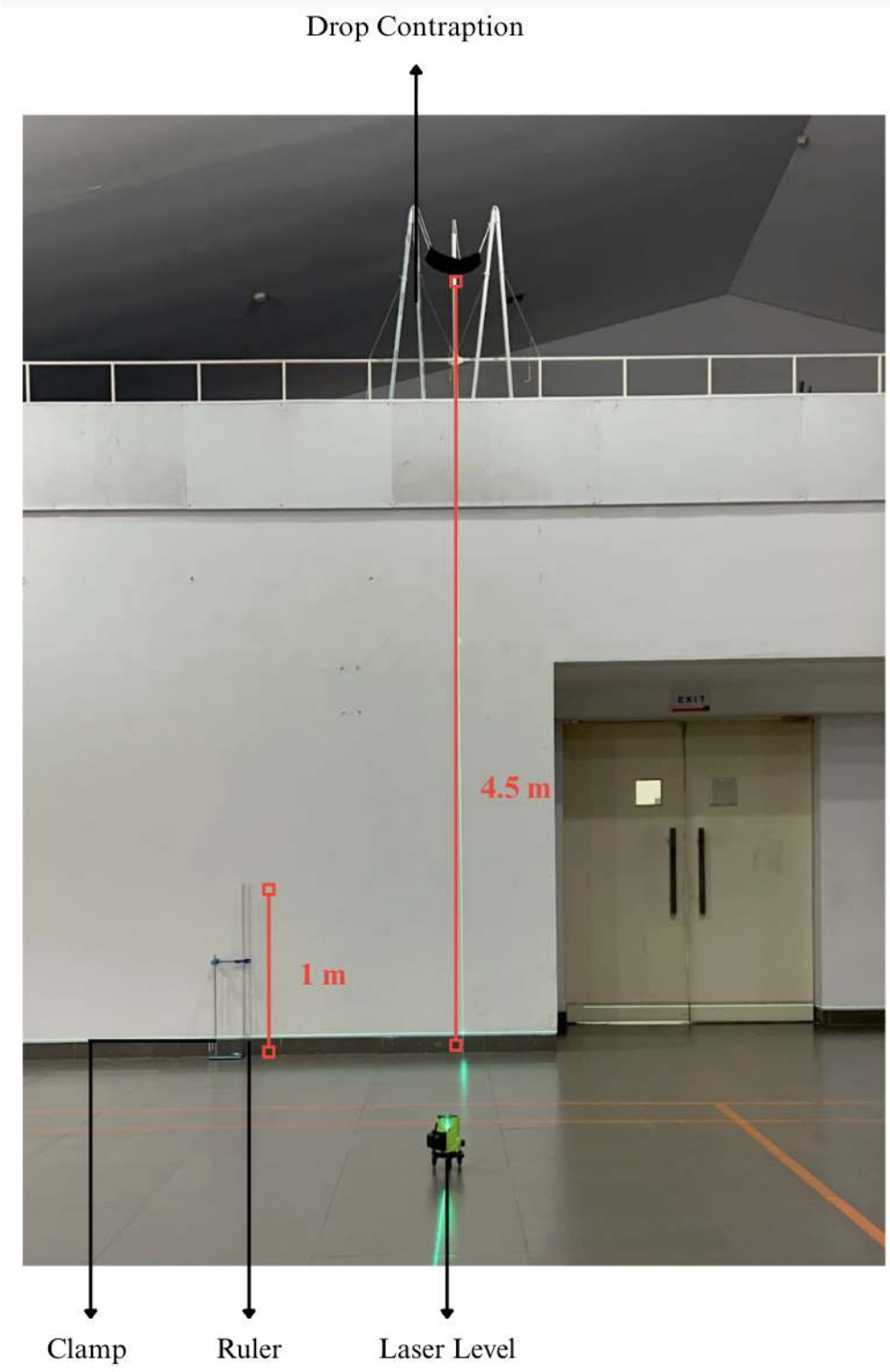


Figure 4: Practical setup of the experiment (annotated with Canva)

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Figure 5: Practical setup of the drop contraption to drop the object (annotated with Canva)

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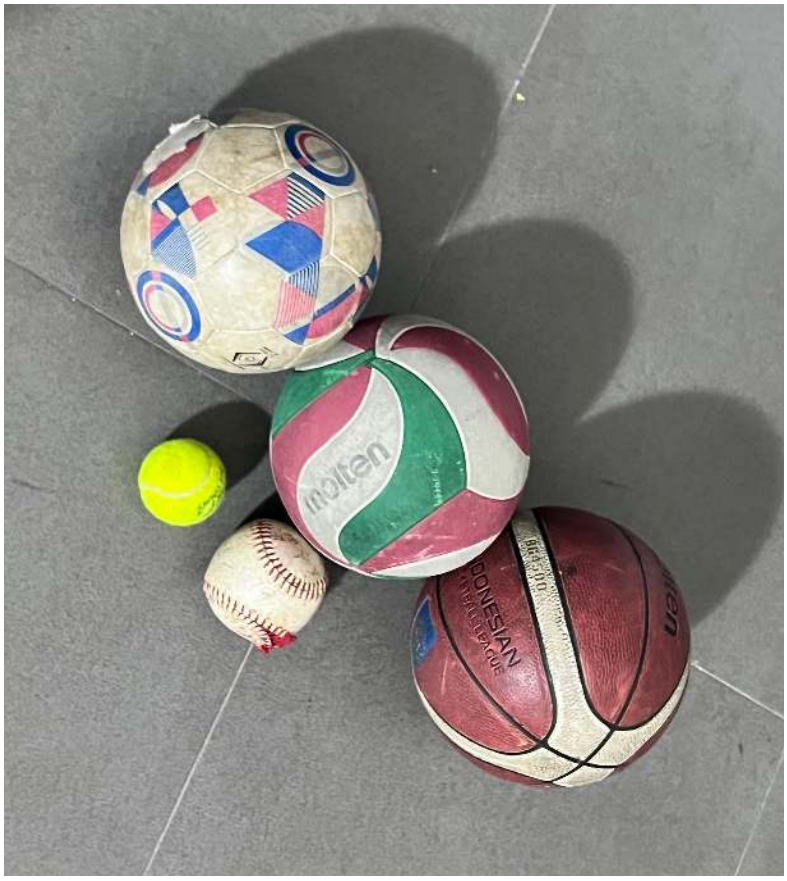


Figure 6: Selected objects

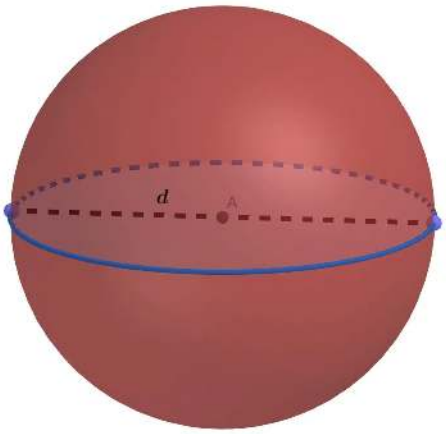


Figure 7: The cross-section taken at the widest point of the sphere (Sherpas)

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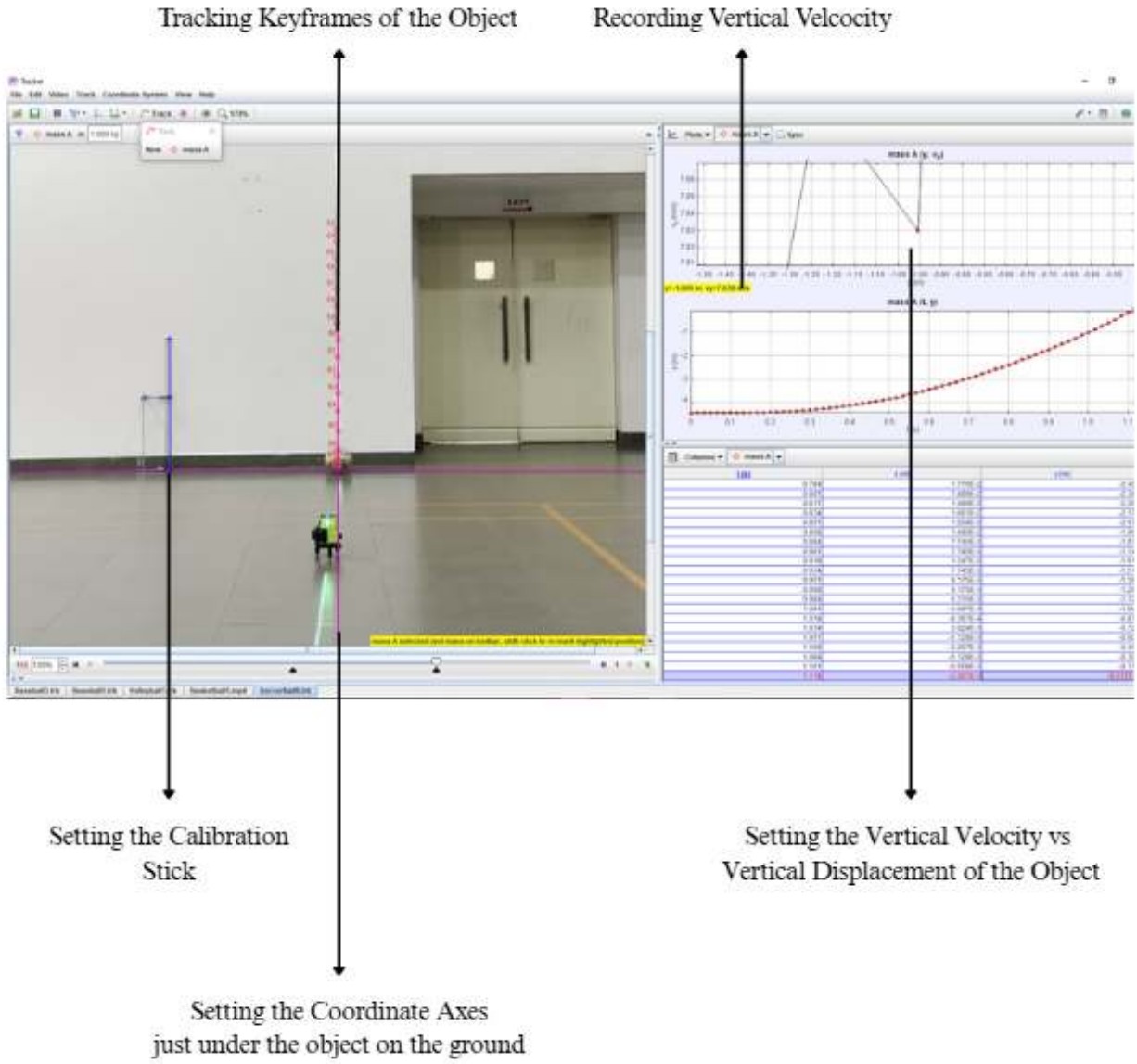


Figure 8: Preview of Tracker Application Setup (annotated with Canva)

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1.7.2 Procedures

Measuring Surface Area-To-Mass Ratio

1. Measure and record the mass of each spherical object using the digital scale.
2. Measure and record the circumference of the cross-section of each spherical object with a rope and measuring tape, by wrapping the rope around the cross-section and measuring its length **(Figure 7)**.

Measuring Final Velocity

3. Calculate the radius of each sphere and find the surface area of each sphere by using **Equation (6)**.
4. Calculate and record the surface area-to-mass ratio using **Equation (5)**.
5. Set up the camera, calibration stick, laser grid, and dropping mechanism.
6. Choose 1 object, place the ball on the dropping mechanism, and measure its height from the ground, ensuring it is 4.500 meters.
7. Start the recording and drop the ball.
8. Stop the recording.
9. Upload the video to Tracker, set up the calibration stick to be 1.000 meters on the ruler on the clamp, and the coordinate axes just under the object on the ground.
10. Plot the graph as the vertical velocity (ms^{-1}) against vertical displacement (m) **(Figure 8)**.
11. Track the object by setting keyframes for each frame.
12. Record the value of the vertical velocity when the vertical displacement is -1.000 meters.
13. Repeat steps 8-14 four more times. 5 trials is used due to constraints in time and reliability in addressing random errors.
14. Repeat steps 8-15 for the remaining four objects.

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1.8 Risk Assessments

Type of Consideration	Description
Safety Considerations	All balls used are lightweight and are distanced from surrounding equipment. The area is clear of any living organisms.
Ethical Considerations	No living organisms are used, hence no ethical considerations.
Environmental Considerations	No chemicals or hazardous substances are involved; waste is minimized.

Table 3: Risk Assessments

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2. Analysis

2.1 Selection of Test Values

The selection of my objects considers the intervals between the calculated surface area-to-mass ratios. When measuring the radius (r) of a sphere from its cross-sectional circumference, we can utilize the formula:

$$r = \frac{circumference}{2\pi}$$

Since the uncertainty of the cross-sectional area is equal to the uncertainty of the measuring tape $\pm 0.0005m$, the uncertainty of the radius (Δr) is:

$$\Delta r = \frac{0.0005}{2\pi} \approx 0.00008m \ (1\ s.f.)$$

Then, the surface area (A_s) and its uncertainty is calculated using **Equation (6)**. A sample calculation is done for the baseball:

$$\frac{\Delta A_s}{A_s} = |2| * \frac{\Delta r}{r}$$

$$A_s = 4\pi(0.04545)^2 = 0.02596 \ (5\ d.p.)$$

$$\Delta A_s = 0.0256(\frac{2*0.00008}{0.04545}) \approx 0.00009m^2 \ (1\ s.f.)$$

This means that the calculated surface area-to-mass ratio (k) also differs. A sample calculation is done for the baseball:

$$\frac{\Delta k}{k} = \frac{\Delta A_s}{A_s} + \frac{\Delta m}{m}$$

$$k = \frac{0.02596}{0.17145} = 0.1514 \ (4\ d.p.)$$

$$\Delta k = 0.1514(\frac{0.00009}{0.02596} + \frac{0.00001}{0.17145}) \approx 0.0005 \ (1\ s, f.)$$

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Measurement of Selected Objects					
Objects	**Circumference** $circumference/m$ $\pm 0.0005\ m$	**Radius** r/m $\pm 0.00008\ m$	**Surface Area** A_s/m^2 $\Delta A_s/m^2$	**Mass** m/kg $\pm 0.00001\ kg$	**Ratio** k/m^2kg^{-1} $\Delta k/m^2kg^{-1}$
Baseball	0.2850	0.04545	0.02596 ±0.0009	0.17145	0.1514 ±0.0008
Tennis Ball	0.1970	0.03135	0.01235 ±0.00010	0.05773	0.2139 ±0.0018
Basketball	0.7430	0.11825	0.17572 ±0.00024	0.54649	0.3215 ±0.0004
Soccerball	0.6320	0.10059	0.12715 ±0.00020	0.30377	0.4185 ±0.0007
Volleyball	0.6760	0.10759	0.14546 ±0.00022	0.26340	0.5522 ±0.0008

Table 4: Surface Area to Mass Ratio of objects used for the experiment

These objects are specifically selected as my range of objects are limited, and hence, I had to select objects that attained approximately consistent intervals.

2.2 Data Collection

2.2.1 Quantitative Data (Theoretical)

The calculation of the theoretical data refers to **Equation (13)**. Since the height of the setup is $4.5m$, and the velocity is being measured 1 meter above the ground, hence take $x = 4.5m - 1m = 3.5m$.

As an example, for baseball, the final vertical velocity is:

$$v_f = \sqrt{\frac{8*9.81}{0.47*1.29*0.1514}} * \sqrt{1 - e^{-\frac{0.47*1.29*0.1514*3.5}{4}}} \approx 8.1231 ms^{-1}\ (4\ d.p.)$$

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Using the usual approach of using operations for calculating uncertainty is too complex. I will resort to the standard deviation approximation, where I approximate the uncertainty based on the maximum possible v_f with the minimum possible v_f of each object. Since we have already found Δk, and v_f is dependent on the inputted value k, I can use the maximum and minimum possible values of k to find the maximum and minimum value of v_f. For the baseball:

$$k_{max} = 0.1514 + 0.000844 = 0.152244$$

$$k_{min} = 0.1514 - 0.000844 = 0.150556$$

When k_{max}:

$$v_f = \sqrt{\frac{8*9.81}{0.47*1.29*0.152244}} * \sqrt{1 - e^{-\frac{0.47*1.29*0.152244*3.5}{4}}} \approx 8.122193646 ms^{-1} \ (v_{min})$$

When k_{min}:

$$v_f = \sqrt{\frac{8*9.81}{0.47*1.29*0.150556}} * \sqrt{1 - e^{-\frac{0.47*1.29*0.150556*3.5}{4}}} \approx 8.123987871 ms^{-1} \ (v_{max})$$

Hence, the uncertainty of baseball is:

$$\Delta v_f = \frac{v_{max} - v_{min}}{2} = \frac{8.123987871 ms^{-1} - 8.122193646 ms^{-1}}{2} \approx 0.0009 ms^{-1} \ (1 \ s.f.)$$

This process is then repeated for the other objects.

Object	**Theoretical Final Vertical Velocity** v_f / ms^{-1}	**Uncertainty** $\Delta v_f / ms^{-1}$
Baseball	8.1231	±0.0009
Tennis Ball	8.0573	±0.0018
Basketball	7.9464	±0.0005
Soccerball	7.8472	±0.0002
Volleyball	7.7151	±0.0002

Table 5: Theoretical final vertical velocities for each object with its uncertainty

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2.2.2 Quantitative Data (Experimental)

In the Tracker application, since v_f is presented in 3 decimal places, the smallest unit of measurement is $0.001ms^{-1}$. Hence:

$$\Delta v_f = \frac{0.001}{2} = 0.0005ms^{-1}$$

Object	**Ratio** k/m^2kg^{-1} $\Delta k/m^2kg^{-1}$	**Final Vertical Velocity of Object** v_f/ms^{-1} $\pm 0.0005ms^{-1}$				
		Trial 1	Trial 2	Trial 3	Trial 4	Trial 5
Baseball	0.1514 ±0.0008	8.5980	8.5890	8.7290	8.6310	8.4120
Tennis Ball	0.2139 ±0.0018	8.2490	8.3410	8.3970	8.2010	8.0910
Basketball	0.3215 ±0.0004	7.9120	7.8910	8.0640	7.9980	8.0210
Soccerball	0.4185 ±0.0007	7.7300	7.8760	7.9160	7.7590	7.8110
Volleyball	0.5522 ±0.0008	7.5060	7.6220	7.5570	7.5640	7.6540

Table 6: Recorded final velocities for each object

2.2.3 Qualitative Data

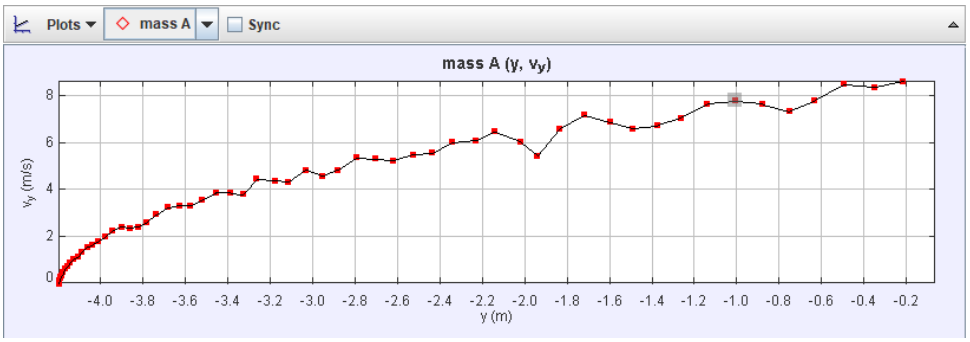


Figure 8: Data analysis (Graph obtained from Tracker) of baseball trial 1

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The qualitative patterns perfectly align with the theoretical model of an object falling with air resistance, with a decreasing slope indicating that acceleration is continuously slowing down. It must be noted that some points in the graph appear jumpy, which may be due to errors in manually tracking the center of mass of the object. This will be acknowledged as a possible source of error.

2.3 Data Processing

To process the experimental data, I will find the mean v_f for each trial and propagate its uncertainty using the standard deviation approximation. I will take one example for each process and repeat it for all other objects. First off, I will take reference from baseball to calculate its mean v_f, denoted as μ_{v_f}:

$$\mu_{v_f} = \frac{v_{trial\,1}+v_{trial\,2}+v_{trial\,3}+v_{trial\,4}+v_{trial\,5}}{5} = \frac{8.598+8.589+8.729+8.631+8.412}{5} = 8.5918 ms^{-1}\ (4\ d.p.)$$

Then, to calculate the uncertainty of the mean, I decided to use the standard deviation approximation as it accounts for random error instead of simply relying on the uncertainty of the instrument:

$$\Delta\mu_{v_f} = 8.5918\left(\frac{5*0.0005}{8.598+8.589+8.729+8.631+8.412}\right) = 0.0005\ (1\ s.f.)$$

Object	**Mean Final Vertical Velocity** μ_{v_f} / ms^{-1}	**Uncertainty** $\Delta\mu_{v_f} / ms^{-1}$
Baseball	8.592	±0.159
Tennis Ball	8.256	±0.153
Basketball	7.977	±0.087
Soccerball	7.818	±0.093
Volleyball	7.581	±0.074

Table 7: Processed data table for mean final vertical velocity

Now, I will take both values and plot them into a graph:

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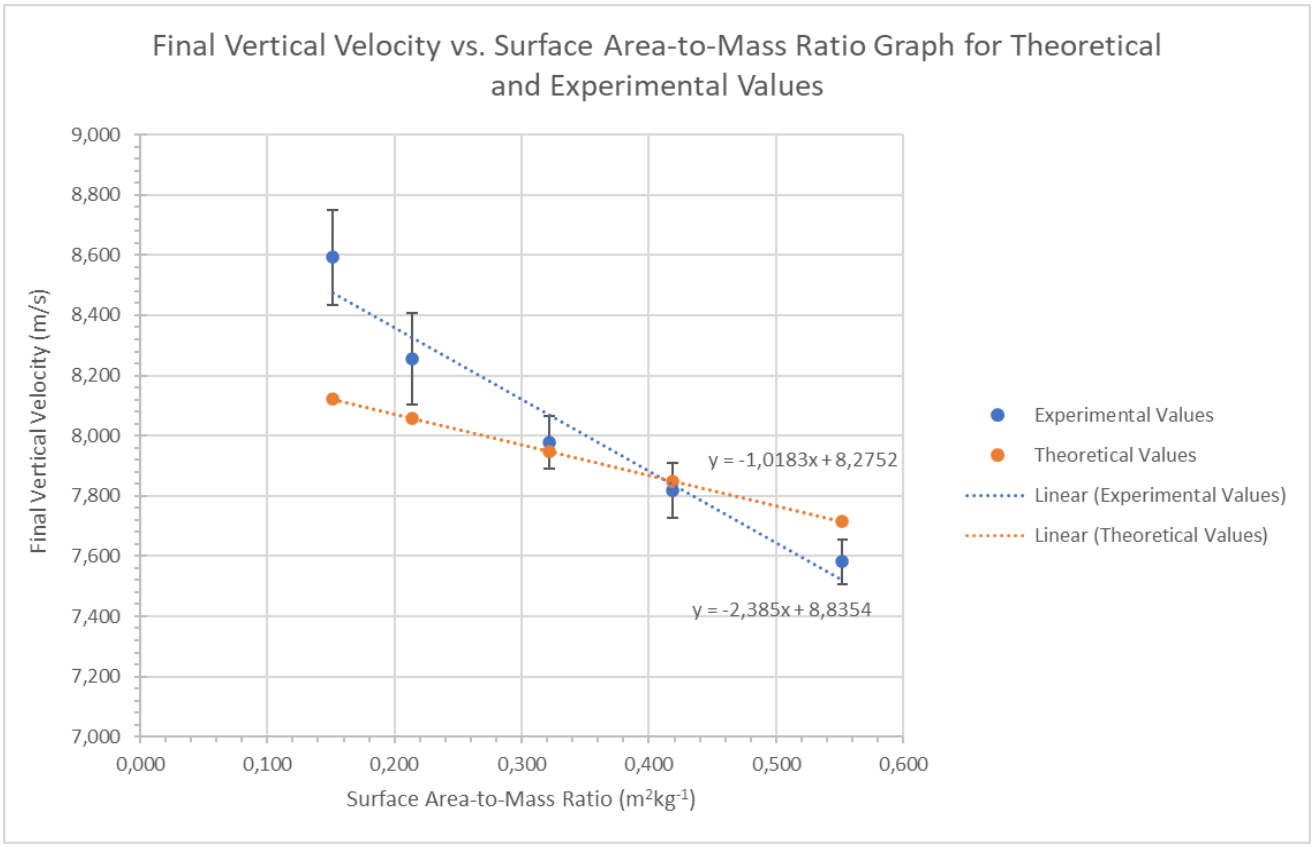


Figure 9: Graph of final vertical velocity vs surface area-to-mass ratio for theoretical and experimental (graphed with Microsoft Excel)

It must be noted that since the values of most error propagations are small, they are not readily visible in the graph. Additionally, the y-axis begins from 7.000 because this allows for greater visual distinction between the data points and their corresponding vertical error bars.

Both models generally show that the v_f decreases as the k increases. However, the rate of decrease of v_f in the experimental model is much greater than in the theoretical model, reflected in its line of best fit, where the gradient is -2.385 as opposed to the theoretical value's -1.0183. This shows that the actual drag force is greater than what the model accounted for. Looking at the uncertainty bars, the theoretical line does not pass through the error bars of the experimental points for most of the data, indicating a systematic difference between the models.

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Due to the exponential nature of its relationship, I will linearize the graph of the experimental value to try to create a line of best fit and find Pearson's coefficient. Since the relationship between the two variables is $v_s \propto \frac{1}{\sqrt{k}}$, I will plot v_f versus $\sqrt{k}$. A sample calculation is done below for the baseball:

$$k_{linearized} = \sqrt{k} = \sqrt{0.1514} = 0.3891015292 \approx 0.3891\ (4\ d.p.)$$

The uncertainty must also be propagated. A sample calculation is done for the baseball:

$$\frac{\Delta k_{linearized}}{k_{linearized}} = \frac{1}{2} * \frac{\Delta k}{k}$$

$$\Delta k_{linearized} = \frac{1}{2} * \frac{\Delta k}{k} * k_{linearized}$$

$$\Delta k_{linearized} = \frac{1}{2} * \frac{0.000844}{0.1514} * 0.3891 \approx 0.0011\ (4\ d.p.\ for\ consistency\ with\ other\ values)$$

This process is repeated:

Object	**Linearized Ratio** $k_{linearized}/m^{-2}kg^{-1}$	**Uncertainty** $\Delta k_{linearized}/m^{-2}kg^{-1}$
Baseball	0.3891	0.0011
Tennis Ball	0.4625	0.0019
Basketball	0.5670	0.0003
Soccerball	0.6469	0.0005
Volleyball	0.7431	0.0006

Table 8: Linearized ratios and their corresponding uncertainties

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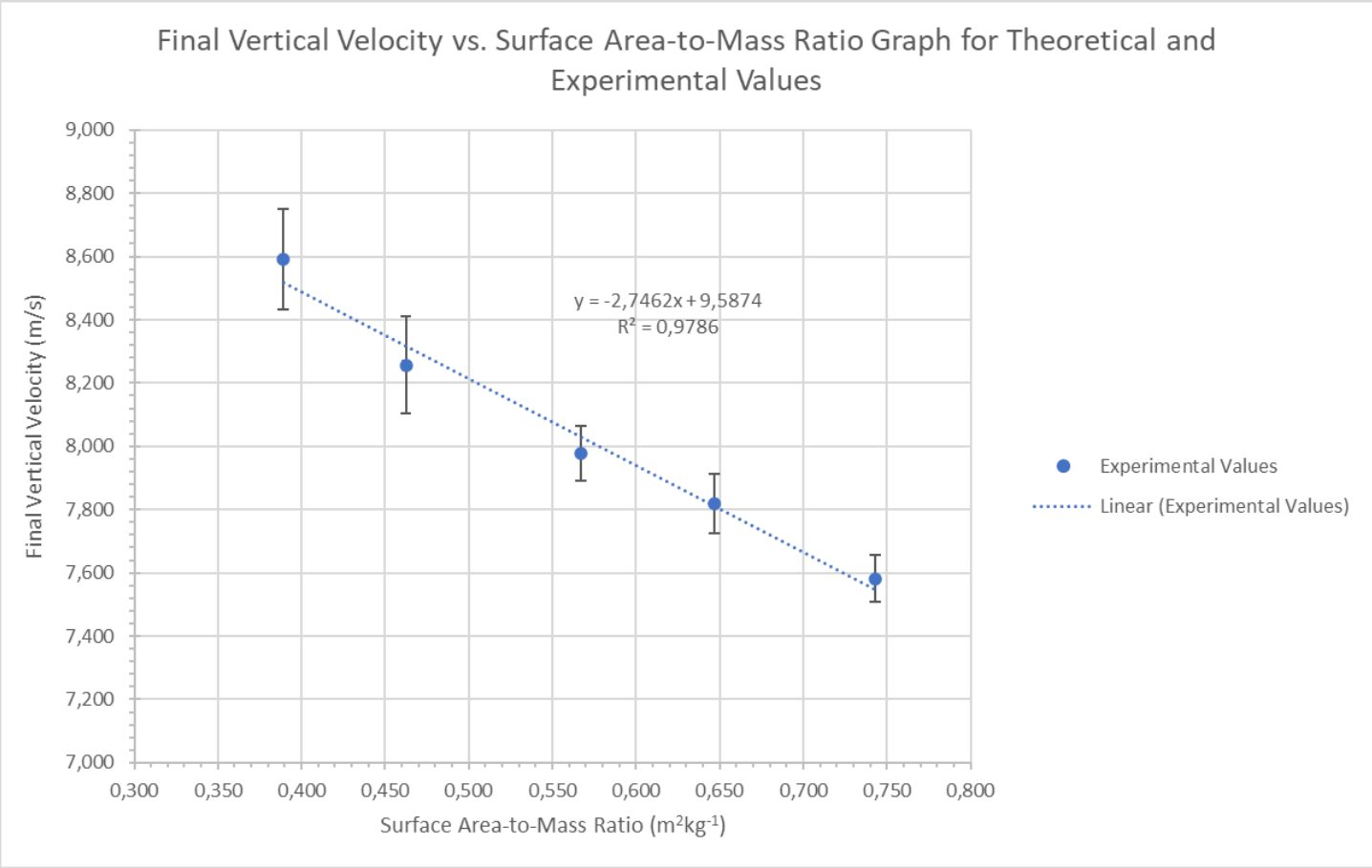


Figure 10: Linearized graph of final vertical velocity vs surface area-to-mass ratio (graphed with Microsoft Excel)

From this, we can find Pearson's coefficient (R^2), a statistical measure that evaluates the strength of the relationship between two continuous variables (Turney, 2022). From the graph, it is 0.9892. This suggests a strong negative linear correlation between the two variables. However, minor deviations from the line of best fit suggest that the relationship is more complex than an exponential pattern. Additionally, I will be calculating the p-value, the likelihood of obtaining the observed data under the null hypothesis of a statistical test, where a p-value less than 0.05 is considered to have the null hypothesis rejected (Beers).

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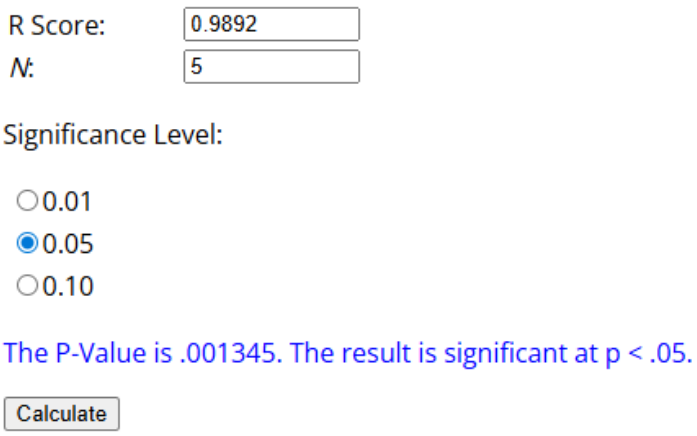


Figure 11: Calculation of the p-value of untreated and treated rooms (Social Science Statistics, n.d.)

Using an online calculator, with R as the Pearson correlation coefficient and N as the sample size, I found the p-value to be 0.001345, which is less than 0.05. Therefore, the null hypothesis can be rejected. Then, I will calculate the minimum and maximum slope. This is done by calculating the minimum and maximum points on the left and right-hand points of the data set, plotting it in Excel, and displaying the equation. Since the uncertainties of the ratios are insignificant (as their error bars are barely visible), I will neglect them and focus on the velocities:

$$v_{f\ max\ LHS} = 8.592 + 0.159 = 8.751$$

$$v_{f\ min\ LHS} = 8.592 - 0.159 = 8.433$$

$$v_{f\ max\ RHS} = 7.581 + 0.074 = 7.655$$

$$v_{f\ min\ RHS} = 7.581 - 0.074 = 7.507$$

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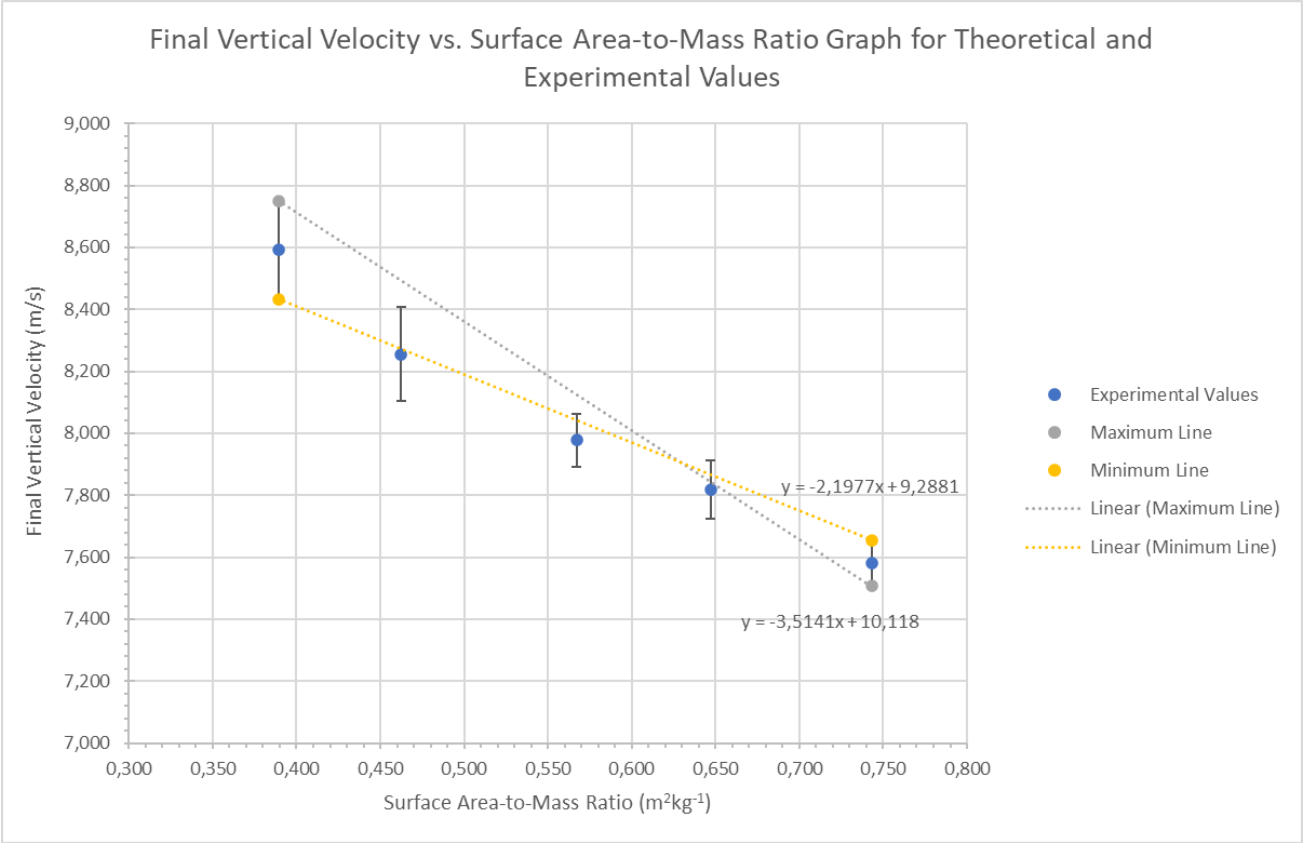


Figure 12: Minimum and maximum lines (graphed with Microsoft Excel)

These points result in a minimum gradient of -2.1977 and a maximum gradient of -3.5141. We can use this to calculate the percentage uncertainty of the error propagation itself, by comparing both values with the line of best fit gradient:

$$Percentage\ Uncertainty\ Max\ Gradient = \left|\frac{-3.5141-(-2.7462)}{-2.7462}\right| * 100\% = 28.0\%\ (3\ s.f.)$$

$$Percentage\ Uncertainty\ Min\ Gradient = \left|\frac{-2.1977-(-2.7462)}{-2.7462}\right| * 100\% = 20.0\%\ (3\ s.f.)$$

A percentage uncertainty of 20% to 28% in the calculated gradient is quite high and signifies low precision, despite having formed a linear pattern from linearization. This shows that the falling of an object is a result of a multitude of factors, leading to significant random scatter in the individual final velocity measurements. Nevertheless, this high uncertainty does not negate the primary finding of a systematic error, as the entire plausible range of experimental slopes remains significantly steeper and far removed from the theoretical slope of -1.0183. This high percentage uncertainty should be addressed in future extensions.

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3. Conclusion

This investigation successfully supported the hypothesis that v_f is inversely proportional to the square root of k. The linearization, which yielded a straight line $y = -2.7462x + 9.5874$, has a strong Pearson's coefficient of 0.9892, confirming this. This was further validated by a p-value of 0.001345, allowing the null hypothesis to be rejected. However, the experimental data systematically deviates from the theoretical predictions, shown through the experimental gradient (-2.385) being significantly steeper than the theoretical gradient (-1.0183), indicating that the real-world objects are more affected by the drag force than the model predicts. This is reflected based on a computational study by IMPULSE, investigating the effect of air friction on free-falling objects, which showed that the velocities of different objects resulted in a significant difference in the maximum velocity and time when compared to idealized models (Miyaqi et al., 2023, 39-52).

The percentage uncertainty formed from the minimum and maximum lines is 20% and 28%, respectively, signifying low precision of the experimental data, showing that my experimental process still resulted in many random errors. This may be because of the standard deviation approximation, which does not directly reference the uncertainty of the Tracker application itself. This highlights the importance of controlling random errors in the experimental procedure in the future. Furthermore, it can be seen that the highest uncertainties occurred at lower k values, especially in experimental results. This may be because theoretical and experimental values showed the greatest systematic difference at this point. Therefore, the increased uncertainty at low k speeds strengthens the conclusion that the theoretical model is most inaccurate in predicting the v_f for objects with low k values.

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4. Evaluation

4.1 Errors and Inaccuracies

Errors and Inaccuracies		
Type of Error	**Description**	**Possible Solution**
Systematic Error	**Calibration/Scaling Error** If the calibration stick was not perfectly vertical or was not exactly 1 meter, all distance measurements would not be precise.	Position the camera as far back as possible while still clearly capturing the motion to prevent parallax error. Ensure the camera is parallel to the wall.
	Non-Uniform Objects The assumption that all objects are perfect spheres with uniform density leads to an inaccurate calculated k and the drag force.	Acknowledged as a limiting factor. Used everyday objects which are widely accepted as spherical.
Random Error	**Starting Condition Variation** Inconsistent release angle or slight initial lateral velocity when dropping the ball introduces random components to the final velocity measurement.	Establishing a new ball drop method that utilizes a consistent release angle and zero net force.
	Statistical Fluctuations The data recorded in the software appears jumpy, affecting the slope calculation in an unpredictable way.	Acknowledged as a limiting factor. Manual tracking and experimenting on a bright surface have been done.
	Measurement of Circumference Measuring the cross-section circumference with a rope and measuring tape introduces random error in aligning the rope and reading the tape.	Marking the starting position of the rope measurement to know precisely when 1 full revolution has elapsed for accuracy. Cutting the rope after alignment.

Table 9: Analysis of possible errors and inaccuracies during experimentation

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4.2 Extension

By applying the proposed possible solutions to future experiments, the experiment may benefit from significantly increased precision due to reduced random error, enabling a more reliable determination of the experimental gradient, and will gain greater physical validity by addressing the systemic errors arising from imperfect object geometry and the failure of the simplified theoretical drag model.

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5. Bibliography

Beers, B. (n.d.). *P-Value: What It Is, How to Calculate It, and Examples*. Investopedia. Retrieved October 26, 2025, from https://www.investopedia.com/terms/p/p-value.asp

Benson, T. (n.d.). *Shape Effects on Drag*. NASA. Retrieved October 13, 2025, from https://www.grc.nasa.gov/www/k-12/VirtualAero/BottleRocket/airplane/shaped.html

Miyaqi, I. A., Supardi, & Barokah3, N. (2023, November 20). Computational Analysis for Free Fall Cases with Air Friction. *IMPULSE: Journal of Research and Innovation in Physics Education*, *3*(1), 39-52. file:///C:/Users/User/Downloads/impulse-7717-Article+Text-25515.pdf

National Science Foundation. (n.d.). *Gravity on Earth*. National Science Foundation. Retrieved October 13, 2025, from https://nsf-gov-resources.nsf.gov/files/Gravity.pdf

PK, S. (n.d.). Vernier Callipers Guide. In *Experimental Physics* (p. 34). Retrieved October 14, 2025, from https://www.scribd.com/document/666981352/Vernier1

Shah, G. (n.d.). *Density of Air: Formula, Units & Applications Explained*. Vedantu. Retrieved October 14, 2025, from https://www.vedantu.com/physics/density-of-air

Sherpas. (2023, November 28). *Volume d'une boule & aire d'une sphère {géométrie} - Sherpas*. Les Sherpas. Retrieved October 14, 2025, from https://sherpas.com/blog/volume-boule-aire-sphere/

SMU. (2009). *Drag*. s2.smu.edu. Retrieved October 1, 2025, from https://s2.smu.edu/propulsion/Pages/dragmain.htm

Social Science Statistics. (n.d.). *Quick P Value from Pearson (R) Score Calculator*. Social Statistics. Retrieved October 26, 2025, from https://www.socscistatistics.com/pvalues/pearsondistribution.aspx

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Tsokos, K. A. (2023). *Physics for the IB Diploma Coursebook with Digital Access (2 Years)*. Cambridge University Press. Retrieved September 30, 2025, from https://www.scribd.com/document/708195178/Physics-K-a-Tsokos-Seventh-Edition-Cambridge-2023-1

Turney, S. (2022, May 13). *Pearson Correlation Coefficient (r) | Guide & Examples*. Scribbr. Retrieved October 26, 2025, from https://www.scribbr.com/statistics/pearson-correlation-coefficient/

Wise. (n.d.). *Terminal Velocity*. Miss Wise's Physics Site. Retrieved September 30, 2025, from https://misswise.weebly.com/terminal-velocity.html